Debris Avalanche and Sector Collapse

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1 Introduction

The idea behind this exercise is to implement some highly simplified models for volcano sector collapse in your Jupyter Notebook on Victor. The problem of volcano sector collapse is laid out in some detail in the following – with simplified equations finding the area of inundation by sector collapse and whether an area is safe or unsafe given a sector collapse.

Experiment with adding text and images to your notebook (like the background material). Then implement equations to describe the sector collapse. You can use the simplified statistical model in the following, or come up with a completely differnt approach!

2 Background

A debris avalanche is a volcaniclastic water-unsaturated gravity-driven flow that results from the sudden partial collapse of the unstable flanks of a volcanic edifice. Debris avalanches are composed of a chaotic, but locally coherent, mixture of rocks, soil and debris mixed with only minor amounts of liquid water, vapor, or ice. Debris avalanches can occur at volcanoes that have no signs of unrest, or might have been inactive for long periods of time, during eruptions that do not directly involve magma, as well as during magmatic eruptions. Their formation is not necessarily dependent on having eruptive activity. Debris avalanches can travel several tens of kilometers before they stop. In contrast to laharic flows, which are water saturated, debris avalanches are water-unsaturated flows, in which particles are maintained in suspension and transported as a result of viscous grain-grain interactions rather than suspension in water. Hence, solid grain forces dominate the physics of avalanches (Iverson, 1997). As the moving debris rushes down a volcano and into river valleys, it incorporates water, snow, trees, buildings, and anything else in the way.

Debris avalanches form a debris avalanche deposit (DAD) that consists of a chaotic mixture of coherent blocks of varying sizes (up to decameters) surrounded by a matrix. The debris avalanche block facies consists of large fractured and deformed fragments mostly derived from the source volcano, that preserve original internal structures and textures. The debris avalanche matrix facies consists of a mixture of smaller volcanic fragments derived from various parts of the source volcano and mixed with a minor amount of exotic materials. DADs often can be traced to a horse-shoe shaped steep-walled structure representing the surface along which part of the edifice collapsed. DADs are characterized by an irregular hummocky topography consisting of numerous overlapping irregular-shaped to conical and rounded hills rich in blocks separated by low-lying areas and depressions. Characteristic textural features of DADs include the presence of: 1) deformed, stretched yet coherent avalanche blocks of varicolored hydrothermally altered material from the former interior of the edifice; and 2) angular fragments of rock that show a diagnostic jigsaw fit across fractures and which form clasts that have remained semi-coherent and have not totally disaggregated and mixed with the finer matrix material despite many kilometers of transport in the avalanche.

Avalanches with volumes greater than 10^6 m^3 are generally of long run-out type. Long run-out avalanches are emplaced in a catastrophic way, with observed or inferred velocities of 20–100 m s^{-1} (Kelfoun and Druitt, 2005). Debris avalanches from composite volcanoes may exceed 10^{10} m^3, and associated deposits can extend more than 100 km from the volcano. Sometimes volcanic avalanches are hot (up to 100 C).

Sector collapses originate debris avalanches, which are characterized by very different velocities of the mobilized mass, ranging from creep-like movements like that at Valle del Bove, Etna Volcano, with velocity of about $10^{-10}\,\mathrm{m\,s{-}1}$ (Froger et al., 2001), to catastrophic fast moving landslides like that at Mount St. Helens in 1980, with velocity of about $10^{2}\,\mathrm{m\,s{-}1}$ (Voight et al., 1981). Sector collapse events might recur just once or twice during the lifespan of a volcano, but in some cases repetitive collapse occurs repeatedly and can even be a characteristic behavior of a particular volcano (eg. Soufrière of Guadeloupe, Komorowski et al., 2005; Komorowski et al., 1997). Volumes associated with flank failures of stratovolcanoes can range from 0.1 to $> 100\,\mathrm{km}^3$, posing a severe hazard both on the edifice flank itself and in downslope areas (e.g., Reid et al. 2000).

Large scale avalanches normally occur on steep volcanoes. Volcano collapse is typically produced by a combination of events rather than a single process (Voight and Elsworth, 1997). They can be triggered by volcanic earthquakes, injection of magma and associated increases of porewater pressure within the volcano and loading by a growing dome (e.g. Soufrière Hills, Montserrat, Voight et al., 2002), weakening of the volcanic edifice by tectonic processes, or chemical alteration of the rocks by hydrothermal or weathering effects (Day 1996; Voight and Elsworth, 1997; Elsworth and Voight, 1999; Reid et al., 2000; Reid, 2004; van Wyk de Vries et al., 2000). It is possible that some collapses might have occurred without an evident triggering mechanism, due to gravitational instability of the edifice. Reid (2004) suggested that heating from remote magma intrusion at depth can generate temporarily elevated pore-fluid pressures that can destabilize the volcano edifice, but diffusion of heat through volcanoes takes years to decades while mechanically-induced porewater increases bleed off within days. Mechanical pressurization induced by intrusions and compression of aquifers is a faster and more effective way to build pore pressure before it can bleed off (Voight and Elsworth, 1997; Elsworth and Voight 1999; Newhall, 2007). For example, at Krafla and Usu volcanoes, observed mechanically-induced porewater pressure increases can be ¿1 MPa, which can rise and dissipate within hours. As shown by the 1980 eruption of Mt. St. Helens, sector collapse resulting from magma intrusion into the edifice also resulted in a large explosive eruption as a pressurized magma intrusion that stagnated at shallow-depth was suddenly decompressed by the collapse into a large laterally-directed explosion or blast (Voight et al., 1981).

Lateral and vertical collapses appear to be less frequent in stratovolcanoes than in basaltic edifices. Gudmundsson (2012) proposed that stratovolcanoes are tougher than basaltic edifices because they function as high-toughness composite structures composed of layers with widely different elastic properties.

Large-volume debris avalanches are common on the submerged flanks of ocean island volcanoes (e.g., Hawaiian Islands, Cape Verde Islands) and steep coastal cliffs can be caused by catastrophic debris avalanches that can even produce tsunamis.

Large-scale debris avalanches, including sector collapse, represent a particular challenge in volcanic hazards assessment. A volcano may have a potential for future events, based on current topography and rock characteristics, yet had not experienced large debris avalanches in the past if such conditions did not occur in the past. Thus, the analysis can rarely assess large-scale avalanche hazards through extrapolation from the record of past events. The assessment must rely on an understanding of the current and near-term condition of the volcanic system, with regards to its potential for future collapse events. For this reason, it is not possible to estimate the likelihood of future events based solely on the geologic record of past events, as only the current state of the edifice is indicative of the potential for a future collapse event. Certainly, external forcing agents (e.g. regional earthquake, extreme rainfall event) must also be considered in the edifice collapse assessment given the existence of unstable parts of the edifice.

3 Simplified modeling approach

Probabilistic assessment of sector collapse and the resulting volcano debris avalanche represents a major challenge in hazard assessment. The source term, volume of material and edifice area that is mobilized during the event, is a critical factor in estimating the potential very long run out of volcano debris avalanches. Furthermore, the conditions that lead to the collapse of volcanoes are incompletely understood. Some volcanoes never experience this type of collapse and rather are characterized by the slow erosion and degradation

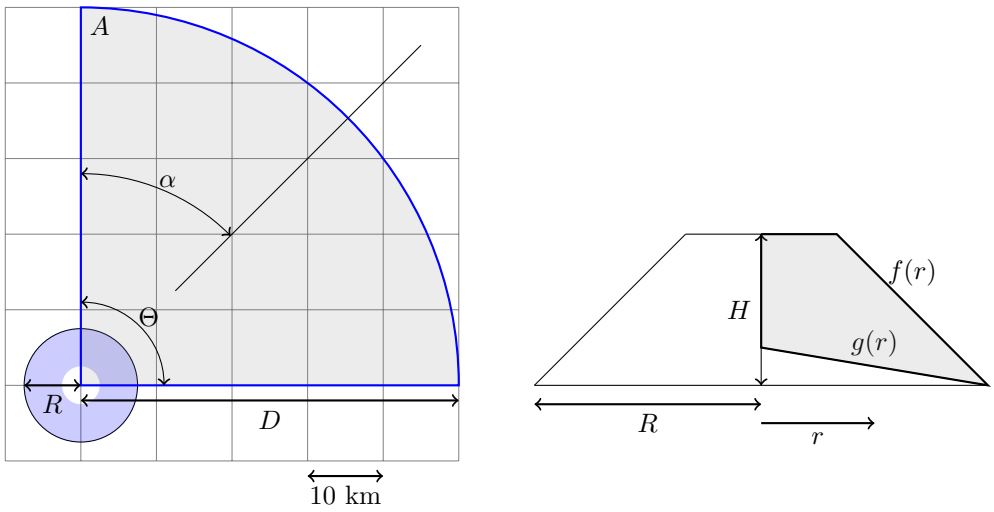


Figure 1: Simplified geometric model of sector collapse and resulting volcano debris avalanche. *Left Panel*: a volcano (shaded circle) of radius R, experiences collapse of a sector described by the central angle, Θ, and azimuth of the centerline of the collapse, α, resulting in inundation by the volcano debris avalanche of an area, A (shaded), up to run out distance, D. Note the map grid is 10 km square, providing an an example of the potential scale of these avalanches. *Right Panel*: In cross-section, the volcano of height, H, collapses. The volume of the sector collapse (shaded) can be modeled by two curves, $f(r)$ describing the original topographic surface, and $g(r)$ describing the new topographic surface after sector collapse. The volume, V, is found by integrating $f(r)$ and $g(r)$ in cylindrical coordinates through the central angle, Θ (see equation 3).

of the cone. Other volcanoes, such as Shivelich (Russia) and Mombacho (Nicaragua) experience multiple collapse events during their life cycles (van Wyk de Vries and Francis, 1997).

In order to assess volcano debris avalanche hazards associated with sector collapse, a conditional probabilistic approach is suggested. The probability of inundation of some location by volcano debris avalanche, given a sector collapse of the volcano, is cast as:

$$\mathrm{P}[\text{inundation}|\text{collapse event}] = f(d, \alpha, \Theta, V) \tag{1}$$

where d is the distance of the location of interest (town, infrastructure) from the center of the volcano, α is the azimuth of the centerline of the volcano debris avalanche, Θ is the central angle of the circle segment (pie-slice) that describes the sector collapse, and V is the volume of the volcano debris avalanche (Figure 1). Equation 1 indicates that the probability of inundation of the location of interest, given a sector collapse event, depends on the distance of the slocation of interest from the volcano, and geometric terms which describe the position and volume of the fraction of the edifice involved in the collapse. Implicitly, additional factors, such as the topography between the volcano and the location of interest, are ignored in this geometric analysis. Nevertheless, because volcanic debris avalanches are highly mobile, and capable of climbing substantial topographic barriers, to a first order the geometric analysis ignoring topography along the avalanche path is warranted. Similarly, involvement of magma in the formation of the volcanic debris avalanche may substantially increase its volume, and the area covered by the resulting deposit. Such factors may be important is some specific hazard evaluations.

The volume of material available for the sector collapse as a function of the central angle, Θ, can be determined for specific volcanoes using a digital elevation model (DEM) of the volcano topography and a model of the fraction of the edifice involved in the sector collapse. Alternatively, the volcano form might be approximated by the change in elevation on the volcano, $f(r)$, as a function of radial distance. For a

cylindrically symmetric volcano, the volume involved in a sector collapse is then:

$$V = \int_0^{\Theta} \int_0^{R} r f(r) \mathrm{d}r \mathrm{d}\theta \tag{2}$$

where R is the radius of the volcano, assuming that entire edifice within Θ is involved in the sector collapse. Alternatively, if $f(r)$ describes the upper surface involved in the collapse, that is, the pre-existing topographic surface, and $g(r)$ describes the lower surface involved in the collapse, that is the post-collapse topographic surface (Figure 1), then the volume is given by:

$$V = \int_0^{\Theta} \int_0^{R} r[f(r) - g(r)] \mathrm{d}r \mathrm{d}\theta. \tag{3}$$

For example, for a a volcano approximated as a perfectly symmetrical cone with height, H and radius, R, a collapse of an entire sector of the edifice, $g(r) = 0$, yields a volume of:

$$V = \int_0^{\Theta} \int_0^{R} r[H(1 - \frac{r}{R}] \mathrm{d}r \mathrm{d}\theta \tag{4}$$

$$= \frac{1}{6} H R^2 \Theta \tag{5}$$

If one-quarter of this conical edifice collapses, as illustrated in Figure 1, then the volume of the volcano debris avalanche is

$$V = \frac{1}{12} \pi H R^2.$$

Given the volume involved in the sector collapse, the next step is to estimate the area covered by the resulting deposit. Like the collapse volume, this area is described in terms of the central angle, Θ (Figure 1). Calder et al. (1999) identified a empirical power law relationship between volume of the sector collapse and the area, A inundated by the flow:

$$A = aV^b \tag{6}$$

In the Calder et al. (1999) model, $b = 2/3$, given the dimensional relationship between volume and area. Using the geological data for cold debris avalanches available in Calder et al. (1999), parameters a and b are estimated as 146 ± 55, and 0.58 ± 0.076, respectively. Note the latter parameter is slightly differnt than $2/3$, but that is ok, likely reflecting our uncertainty. Then the run out distance, D, of the flow is

$$D = \sqrt{\frac{2A}{\Theta}} \tag{7}$$

assuming that the flow spreads laterally and symmetrically. If the azimuth from the volcano to the location of interest lies within the central angle Θ, and if $D > d$, then inundation occurs. For example, Crandell(1989) found that the Mt. Shasta volcano debris avalanche associated with sector collapse of that volcano has a deposit area of $675\,\mathrm{km}^2$, corresponding to a volume of V = $45\,\mathrm{km}^3$. This yields a run out distance of approximately 49 km, the mapped extent of the deposit for a central angle of approximately 0.57 radians (33°). For such a sector collapse event, if the location of interest were located within 49 km of the volcano and on an azimuth within the range described by the central angle, that location would be inundated by such an event. A more detailed analysis is illustrated by Figure 3.

Although this model is geometrically simple, it captures the problem that the main uncertainties associated with sector collapse and volcano debris avalanche are related to the existing condition of the volcano. A probabilistic model depends on estimation of the potential orientation of collapse (azimuth of the centerline of the sector failure; α in Figure 1) and the potential range of central angles. Given a central angle, the volume of the sector collapse can be estimated (equation 3), using current topography to estimate $f(r)$ and assuming a reasonable post-collapse topography, $g(r)$. Given uncertainty in the power law model parameters, a and b, this estimated volume corresponds to a range of potential inundation areas, A and run out

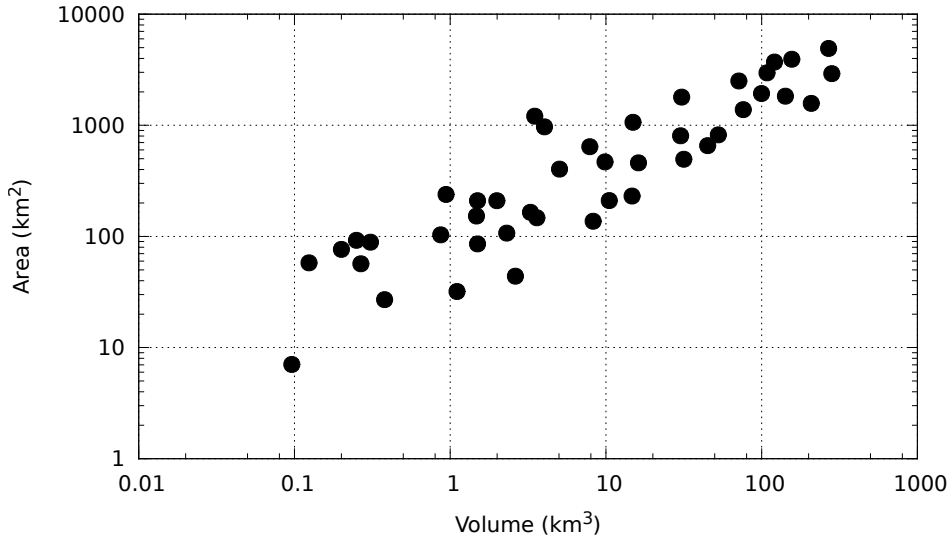


Figure 2: The area inundated by cold volcano debris avalanches as a function of volume for deposits of volumes spanning several orders of magnitude (data from Calder et al., 1999). This relationship can be modeled using a power law, $A = aV^b$, with parameters a and b estimated as 146 ± 55, and 0.58 ± 0.76, respectively, based on these observations from the geologic record (Calder et al., 1999).

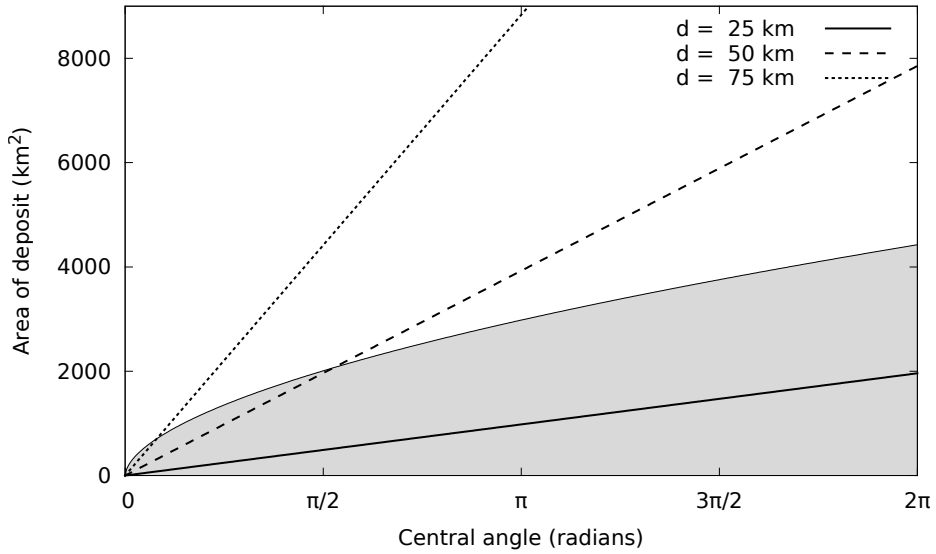


Figure 3: One approach to illustrating the hazard by solving the equations. The deposit area is taken to be a function of central angle, Θ, for a conical volcano, with volume calculated using equation 5, of height $H = 2.4\,\text{km}$ and radius $R = 12.5\,\text{km}$. The deposit area is shown for example run out distances as a function of central angle. Where these lines are within the shaded area, inundation by the volcano debris avalanche is likely. For a location of interest located 25 km from a volcano of this size, virtually any sector collapse would potentially result in inundation of the location of interest. For $d = 75\,\text{km}$, the analysis indicates that the site is too distant from the volcano for inundation, given virtually any sector collapse, and for $d = 50\,\text{km}$ the site is inundated for sector collapse with central angle $\Theta \leq \pi/2$ radians.

distances, D. A Monte Carlo approach might be used to estimate A and D by randomly sampling values for the azimuth (α), central angle (Θ), $g(r)$, a and b.

Geological investigations can be used to constrain the potential ranges of azimuths (α), central angles (Θ), and range of post-collapse topography ($g(r)$) (Reid, 2004). In the simplest approach:

$$\alpha, \Theta, g(r) = f(\text{volcano morphology, alteration, tectonic setting}) \tag{8}$$

Volcano morphology can provide clues about potential azimuth of the centerline of collapse. For example, volcanoes constructed on topographic slopes appear to have a tendency to collapse in the downslope direction of this topographic slope. The edifice of the volcano itself may be steeper on one flank, making it more prone to collapse, or buttressed by older topography on one flank, making it less prone to collapse. The occurrence of alteration zones within the volcano may make a sector more prone to collapse, as might the presence of a fault. Thus, reasonable constrains on model parameters might be deduced from geologic mapping of the volcanoes of interest.

In your Jupyter notebook you might consider alternative topographic models, solving for $f(r)$ and $g(r)$ using more realistic approximations for a specific volcano. You might consider investigation of the affect of uncertainty in the parameters derived by Calder et al. 1999.

Additional Reading

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